Bachelor Project

Czech Technical University in Prague

F3

Faculty of Electrical Engineering
Department of Radioelectonics

Design of a platform for 2D LiDAR

Illia Novosad

Supervisor: Ing. Pavel Puričer, Ph.D.
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Declaration

I hereby declare that artificial intelligence was used during the writing of this thesis, in particular as an aid for generating TikZ figures. All generated content was verified by me. All information taken from the work of other authors has been properly cited.

In Prague, 13. November 2025

Abstract

There are currently many advanced LiDAR sensors available for 3D spatial scanning. However, most of these sensors are too expensive for the average user. The most affordable commercial option is the Unitree L1, which costs $ 250 [1]. The aim of this miniproject is to design a platform for a 360° LiDAR sensor, which will be utilized in a subsequent bachelor's thesis to visualize a 3D image of the scanned environment.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

Supervisor: Ing. Pavel Puričer, Ph.D.

Abstrakt

V současnosti existuje spousta pokročilých lidar-senzorů určených pro 3D snímání prostoru. Většina z nich je však cenově nedostupná pro běžného uživatele. Nejvíc dostupným komerčním řešením je Unitree L1, který stojí $ 250 [1]. Cílem toho miniprojektu je navrhnout pro 360° lidar-senzor platformu a použít ji v navazující bakalářské práci pro vizualizaci 3D-obrazu naskenovaného prostoru.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 2D LiDAR

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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing

SLAM Simultaneous Localization and Mapping

Chapter 1

Introduction

Nowadays, LiDAR sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [2; 3, Ch. 4-6]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the cheapest commercial 3D LiDAR sensor, which was found during the research for this miniproject, costs $249 [1].

The primary motivation behind this mini-project (and the subsequent bachelor's thesis) is to develop a cost-effective platform for a 360° STL-19P LiDAR sensor that can be utilised for 3D spatial scanning and visualisation of the surrounding environment.

Figure 1.1: Photo of STL-19P LiDAR [1]

Table 1.1: Main parameters of STL-19P LiDAR[1]

Typical measuring range [m]	0,03–12
Angular resolution [°]	0,72
Dimensions(WxLxH) [mm, mm, mm]	38,59x38,59x33,50
Communication interface	UART @ 230400 bps

A subsequent bachelor's thesis will then focus on the Digital Signal Processing (DSP) techniques for the effective processing of the acquired point cloud data and proper utilisation of open-source visualisation tools (such as Python's Open3D library [4]) to recreate a 3D scene.

[1]Photo and parameters were taken from `https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/`

Chapter 2

Design of a Platform

This chapter will firstly go over the existing DIY solutions for 360° LiDAR platforms which the author has stumbled upon during the research[too informal?], analyzing their pros and cons. After that, the proposed design will be presented, along with the chosen design decisions and their justifications.

2.1 Existing DIY Solutions for Environmental Mapping

2.1.1 Using a Single-Point LiDAR module

One of the most straightforward approaches to achieve 360° environmental mapping is to utilize a single-point LiDAR module mounted on a platform which is able to rotate and tilt. The implementation could be found in this project: [5].

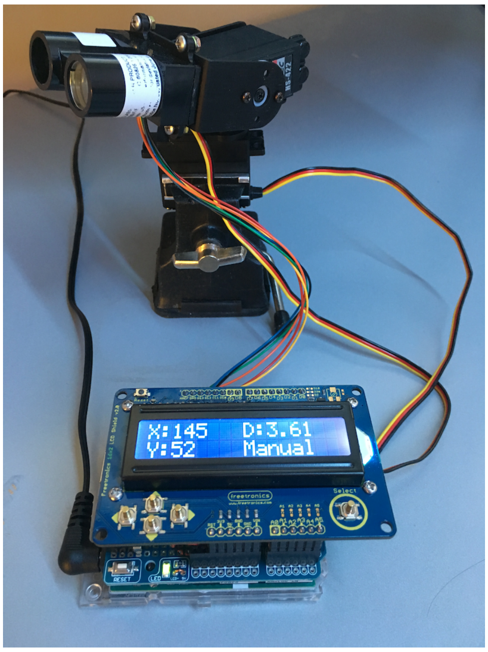


Figure 2.1: Implementation in project [5] using a single-point LiDAR

Pros	Cons
+ Single-point LiDARs are the most cost-effective among all existing types	– Slow data acquisition relative to methods leveraging 2D LiDAR's speed
+ Simple design	– Movement of lidar may introduce innacuracies in distance measurements, which need to be compensated for in code if we want to assume that all points are measured from a single position in space

2.1.2 Using a Single-Point LiDAR and a Mirror

The following approach was used in this project: [6]. A single-point LiDAR is placed below a rotating mirror, which reflects the laser beam horizontally. As the mirror rotates, the laser beam scans the environment in 360°. Moreover, by tilting the mirror at an angle, vertical scanning can also be achieved, allowing for the acquisition of the whole scene.

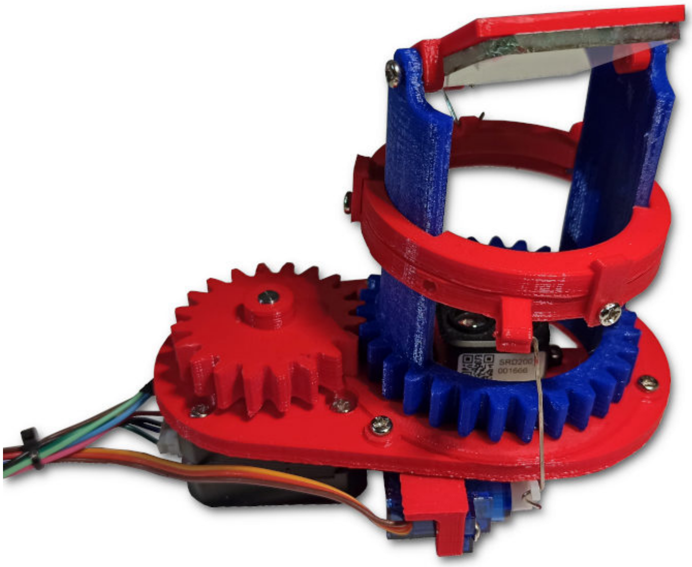


Figure 2.2: Implementation in project [6] using a single-point LiDAR and a mirror

Pros

- + Cost-effectiveness as in previous implementation
- + Simple design
- + Much easier to correct the distance measurement, since the LiDAR is stationary and the distance from it to the mirror is known.

Cons

- – Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- – Mirror's quality may affect the accuracy of distance measurements
- – Mechanical structure holding the mirror may introduce vibrations, thus affecting measurement accuracy

2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform

This approach utilizes a 2D LiDAR sensor mounted on a platform that can tilt the sensor side-to-side, as demonstrated in this project: [7]. The 2D LiDAR scans the environment in a horizontal plane, while the tilting mechanism allows for vertical scanning.

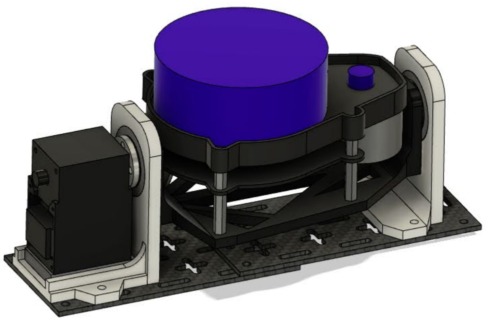


Figure 2.3: Implementation in project [7] using a tilting platform

Pros

+ Faster data acquisition compared to solutions with single-point LiDAR measurements
+ Higher resolution given by the use of a 2D LiDAR sensor
+ This design could be potentially used for Simultaneous Localization and Mapping (SLAM) applications, given the higher speed

Cons

– More expensive than single-point LiDAR solutions
– Point distribution is non-uniform due to the tilting mechanism which produces denser point clouds on the axis of tilting (see fig. 2.4)
– Mechanical design is more prone to vibrations produced by the LiDAR itself affecting measurement accuracy

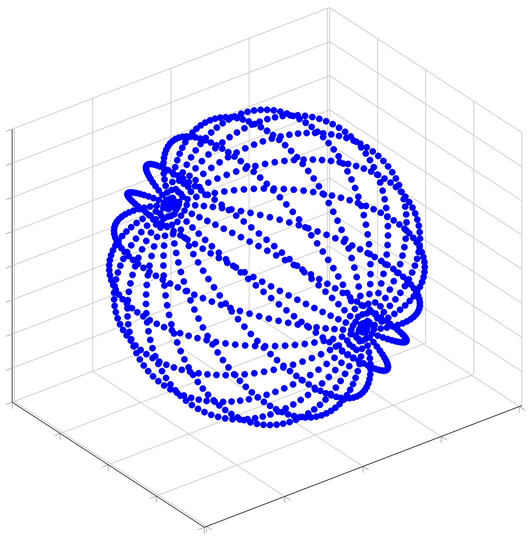


Figure 2.4: Simulation of an obtained point cloud using a tilting platform

2.2 Proposed Design of a 360° LiDAR Platform

Based on the analysis of existing DIY solutions for 360° LiDAR platforms, the proposed design for this miniproject involves using a 2D LiDAR sensor mounted on a tilted platform that is able to rotate around its axis.

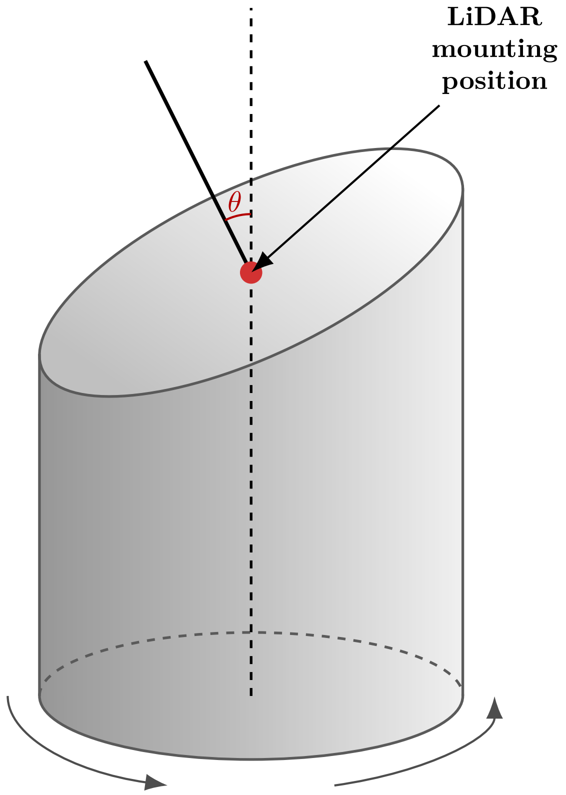
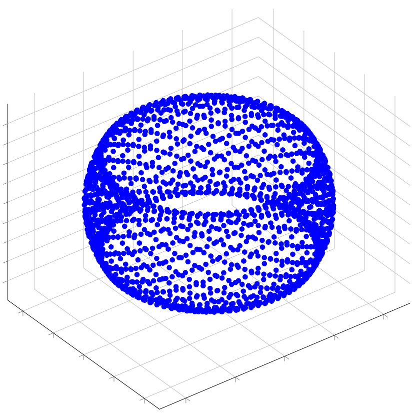


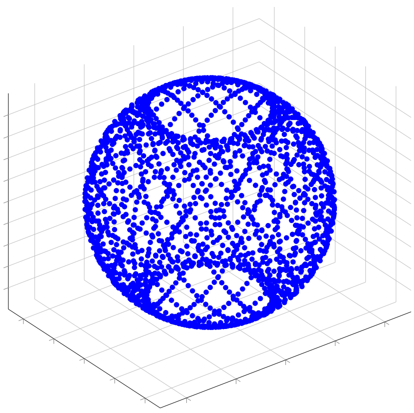
Figure 2.5: Proposed platform design. θ is the tilt angle

This design leverages the advantages of 2D LiDAR sensors, such as fast data acquisition and high resolution, while also eliminating some of the drawbacks of previous designs, namely:

+ The platform will be quite rigid, minimizing vibrations
+ Mechanical design is very simple, making it easy to build and troubleshoot
+ The resulting point cloud will be more uniformly distributed compared to the one obtained using tilting platform design presented in section 2.1.3 unless the tilt angle is extremely steep (see fig. 2.6)



(a) : $\theta = 30^{\circ}$



(b) : $\theta = 60^{\circ}$

Figure 2.6: Simulation of an obtained point cloud using a rotating tilted platform with two different tilt angles θ

Such design does have its drawbacks, however. The first one is immediately apparent from fig. 2.6: there are blind spots in the point cloud where no points are measured. While these blind spots can be minimized by choosing as large a tilt angle as possible, it would reintroduce the problem of non-uniform point cloud density.

Besides that, compared to the tilting platform design presented in section 2.1.3, the proposed design would need to rotate twice as much to obtain all the full coverage of the environment, since the tilting platform scanned only within $[-90^{\circ}, +90^{\circ}]$, whereas the current design must complete a full 360°. This would lead to longer scanning times.

2.2.1 Controlling the Platform

Since we want our platform to be precisely controlled, we need to choose an appropriate motor. Stepper motors are commonly used in applications requiring precise positioning, as they can be controlled to move in small, discrete steps. For this design, a NEMA 17 17HS8401 stepper motor is used to rotate the LiDAR platform.

Figure 2.7: NEMA 17 17HS8401 photo[2]

Table 2.1: Crucial parameters of the chosen NEMA 17 17HS8401 stepper motor[2]

Motor Weight [Ω]	360
Holding Torque [N m]	0,5
Step Angle [°]	1,8
# of Steps per Revolution [-]	200
Rated Current [A]	1,68

For heavy load applications, it is important to consider the potential error sources of stepper motors, such as the stepper motor losing steps due to a heavy load or by driving the motor faster than its maximum speed [8, p. 13]. However, in our case, the load will be relatively light for this motor and the speed will be kept low, thus a feedback mechanism for correcting lost steps is not required.

2.2.2 Choosing the Tilt Angle

The tilt angle of the LiDAR platform is a crucial parameter that affects the size of the aforementioned blind spots in the point cloud (see fig. 2.6). The trade-off for a larger tilt angle, however, is a more spread-out point cloud

[2] Photo and parameters were taken from `https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/`

in the horizontal direction if we rotate our platform in discrete steps [insert figure].

As could be seen in [insert reference to figure], the resolution of the point cloud in the direction perpendicular to the internal rotation axis of the LiDAR is dependent on the tilt angle, specifically

$$\text{res} = \text{step\_angle} \cdot \sin(\theta) \tag{2.1}$$

where θ is the tilt angle and step_angle is constant and is equal to 1.8° for our motor (see table 2.1). A following graph shows how the point cloud resolution changes with the tilt angle:

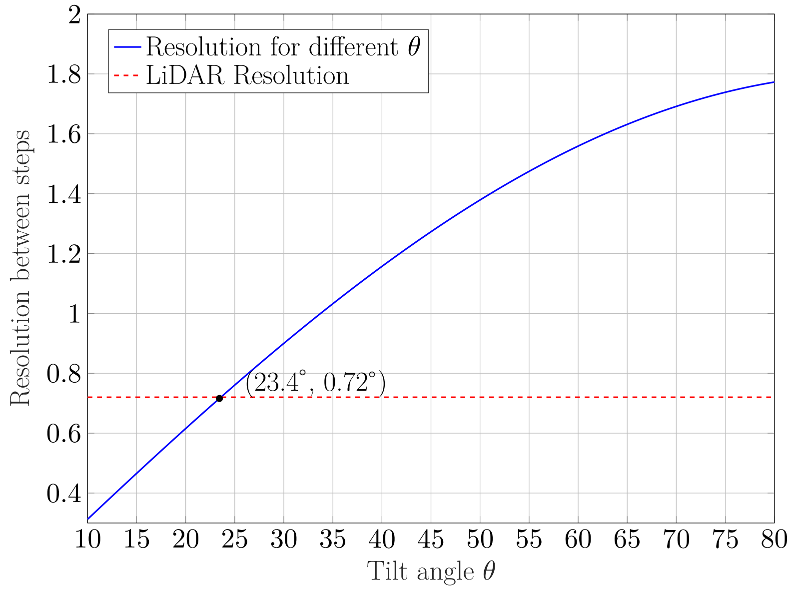


Figure 2.8: Point cloud resolution depending on the tilt angle θ

As could be seen from fig. 2.8, the resolution given by the step angle will be the same as the angular resolution of the LiDAR itself at the tilt angle of approximately 23.4°.

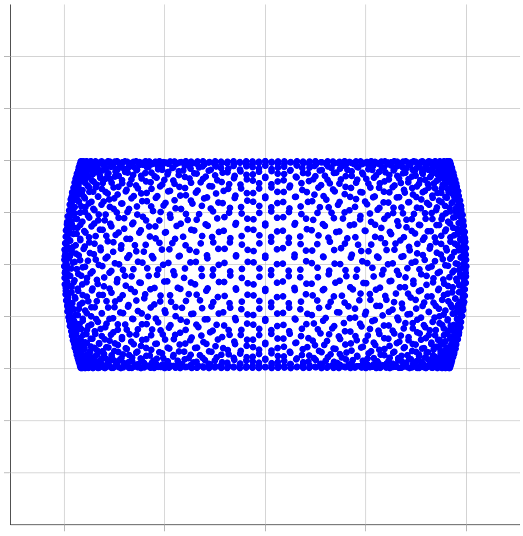


Figure 2.9: Side profile of a point cloud obtained using a rotating tilted platform with tilt angle $\theta = 23.4^{\circ}$

Looking at the side profile of the point cloud obtained using this tilt angle, we can see that the blind spots are extremely large making this tilt angle impractical for our application. Thankfully, there is a way to achieve a more complete coverage of the environment without sacrificing point cloud density.

2.2.3 Choosing a Gear Ratio for the Rotating Platform

In order to remove the need for a compromise between point cloud coverage and density while using a stepper motor to rotate the platform, we can introduce a gear mechanism into our design. Using gears with a specific gear ratios allows us to choose a tilt angle that minimizes blind spots while still maintaining a dense point cloud. The resolution will then be given by

$$\text{res} = \text{gear\_ratio} \cdot \text{step\_angle} \cdot \sin(\theta), \tag{2.2}$$

where gear_ratio is the ratio of the number of teeth on the driven gear to the number of teeth on the driving gear. A following graph is similar to fig. 2.8, but the third dimension represents different gear ratios:

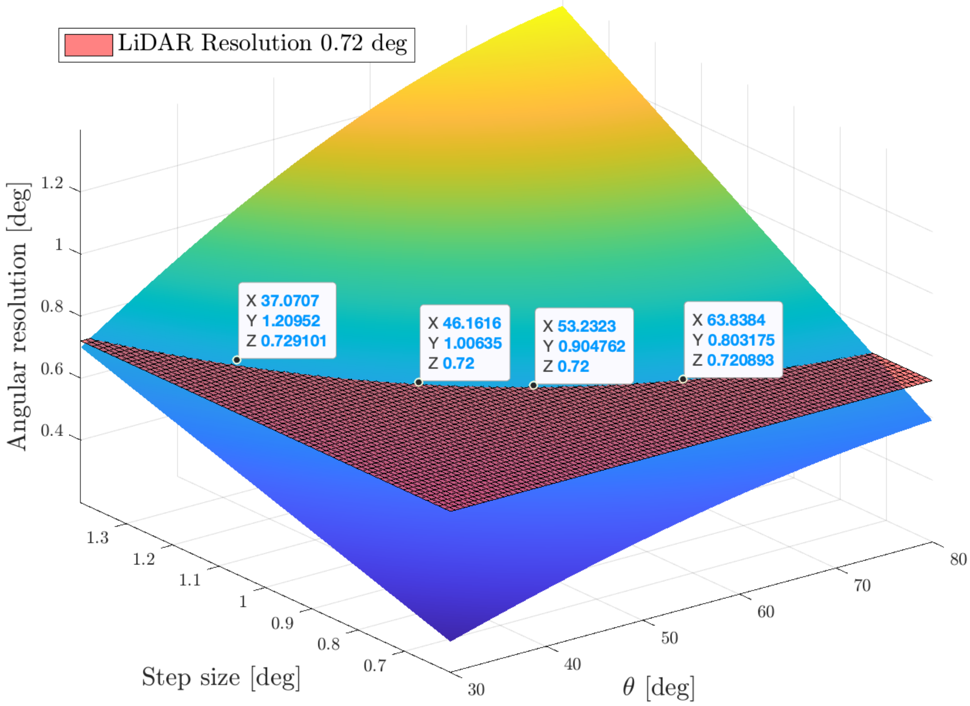


Figure 2.10: Point cloud resolution depending on the tilt angle θ and gear ratio

As could be seen from fig. 2.10, using a gearbox lets us choose from a variety of tilt angles without worsening the angular resolution.

Table 2.2: Best combinations for step angle/gear ratio/tilt angle

Step Angle [°]	Gear Ratio [-]	Tilt Angle [°]
1,2	3:2	37
1,0	9:5	46
0,9	2:1	53
0,8	9:4	64

Of course, by using the gearbox we increase the number of steps needed for the full revolution, thus increasing the scanning time. Nevertheless, by providing platforms of different sizes with appropriate gear ratios, we can make a modular platform which could be reassembled depending on the application

Chapter 3

Conclusion

The main drawbacks of the existing DIY solutions for a cheap LiDAR scanner were presented in section 2.1.

Then, the design of a rotating tilted platform was presented in section 2.2 which solved all but one problem – insufficient resolution.

Introducing a gearbox to our design is a viable solution which will not only lets us scan with higher resolution, but also choose if the scanning speed or point cloud coverage is more important in a certain scenario.

Appendix A

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